

Jiho Shin

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Education

York University, Toronto <i>Dept. of Electrical Engineering and Computer Science</i> Ph.D. Research Field: AI/ML for Software Engineering, Code Generation, Automated Software Testing, LLM for SE Supervisor: Dr. Song Wang and Dr. Hadi Hemmati Dissertation: Investigating the Effectiveness of Large Language Models in Automated Software Engineering	ON, Canada 2021–2025
Handong Global University, Pohang <i>Dept. of Computer Science and Electrical Engineering</i> M.Sc. Research Field: Code Generation, Actionable and Explainable Defect Prediction Supervisor: Dr. Jaechang Nam Thesis: Actionable Defect Prediction	South Korea 2019–2021
Handong Global University, Pohang <i>Dept. of Computer Science and Electrical Engineering</i> B.Sc. Specialization: Web Applications, Human-Computer Interaction, Computer Vision	South Korea 2012–2019

Experience

York University, Toronto <i>Postdoctoral Research Fellow</i> – Conducting and advising graduate researchers on Agentic Software Engineering research.	ON, Canada May 2026–present
Queen's University, Kingston <i>Postdoctoral Research Fellow</i> – Conducting and advising graduate researchers on Agentic Software Engineering studies at SAIL/MCIS Lab.	ON, Canada Sep 2025–Apr 2026
Turing, Palo Alto <i>RLEF/CodeGenAgent Intern (Remote)</i> – Synthesized high-quality Chain-of-Thought code data for reinforcement learning with execution feedback (RLEF). – Improved Code Generation Agents by validating the soundness/completeness of the test generation module.	CA, USA Jan 2025–May 2025
HGS (Hinduja Group Companies), Charlottetown <i>Applied AI Engineering Intern (Remote)</i> – Developed a chatbot for the job application process for prospective workers using <i>Microsoft Copilot Studio</i> . – Implemented another chatbot for the company's homepage to guide prospective clients with <i>lead and sales details</i> .	PE, Canada Oct 2024–Feb 2025
York University, Toronto <i>AI4SE & SE4AI Researcher</i> – AI for Software Engineering (AI4SE): Built agents and LLM frameworks that automate testing, generate code, and repair software using data mined from software repositories, improving software quality and development efficiency. – Software Engineering for AI (SE4AI): Engineered reliable, secure, and interpretable AI systems by testing, validating, and benchmarking LLMs and ML/DL libraries using code repositories, issue trackers, and Q&A sites. <i>Teaching Assistant</i> – Served as teaching assistant and head TA, coordinating graders, office hours, and lab materials (lab tests and lab lecture materials) for <i>Software Design</i> , <i>Software Engineering Testing</i> , and <i>Software Tools</i> courses.	ON, Canada Sep 2021–Aug 2025
Handong Global University, Pohang <i>Software Engineering Researcher</i> – Proposed an actionable defect prediction framework improving the F1 score by 22% over baseline ML models. – Conducted the first survey on 32 studies that generate source code from natural language descriptions . <i>Teaching Assistant</i> – Assisted in grading, developing lecture materials, and holding office hours for <i>Computer Vision</i> , <i>Data Structures</i> , <i>C Programming</i> , and <i>Software Engineering</i> courses.	South Korea Mar 2019–Feb 2021 Sep 2018–Dec 2020

Publications

Conferences

1. **Demystifying Checker Bugs in Deep Learning Libraries**
Harzevili, Mohajer, **Shin**, Wei, Uddin, Yang, Wang, Wang, Jiang, Nagappan, **ICSME 2026** (accept%=27%)
2. **SecVulEval: Benchmarking LLMs for Real-World C/C++ Vulnerability Detection**
MBU Ahmed, NS Harzevili, **Jiho Shin**, HV Pham, S Wang, **Alware 2026 B&D Track**
🏆 Best Benchmark & Dataset Paper Award (1/14=7%)
3. **Retrieval-Augmented Test Generation: How Far Are We?**
Jiho Shin, NS Harzevili, R Aleithan, H Hemmati, S Wang, **ICSE 2026** (accept%: 160/660=24% Cycle 1)
4. **Pre-trained Models for Bytecode Instructions**
DG Kim, TM Kim, **Jiho Shin**, S Wang, HY Choi, JC Nam, **ICST-Short Paper 2025** (accept%: 11/26=42%)
5. **Prompt Engineering or Fine-Tuning: An Empirical Assessment of LLMs for Code**
Jiho Shin, C Tang, T Mohati, M Nayebi, S Wang, H Hemmati, **MSR 2025** (accept%: 44/161=27%)
6. **Domain Adaptation for Code Model-based Unit Test Case Generation**
Jiho Shin, S Hashtroudi, H Hemmati, S Wang, **ISSTA 2024** (accept%: 143/694=21%)
7. **An Empirical Study on the Stability of Explainable Software Defect Prediction**
Jiho Shin, R Aleithan, JC Nam, JJ Wang, NS Harzevili, S Wang, **APSEC 2023** (accept%: 43/128=34%)
🏆 Distinguished Paper Award (3/43=7%)
8. **Automatic Static Vulnerability Detection for Machine Learning Libraries: Are We There Yet?**
NS Harzevili, **Jiho Shin**, JJ Wang, S Wang, N Nagappan, **ISSRE 2023** (accept%: 62/247=25%)
9. **Characterizing and Understanding Software Security Vulnerabilities in Machine Learning Libraries**
NS Harzevili, **Jiho Shin**, JJ Wang, S Wang, N Nagappan, **MSR 2023** (accept%: 43/115=37%)
10. **API Recommendation for Machine Learning Libraries: How Far Are We?**
MS Wei, YC Huang, JJ Wang, **Jiho Shin**, NS Harzevili, S Wang, **ESEC/FSE 2022** (accept%: 99/469=21%)
11. **Similar Patch Recommendation for Actionable Defect Prediction**
Jiho Shin and JC Nam, **KCSE-Short Paper 2020**
🏆 Distinguished Paper Award (3/32=9%)

Journals

1. **StaAgent: An Agentic Framework for Testing Static Analyzers**
E Nnorom, MBU Ahmed, **Jiho Shin**, HV Pham, S Wang, **TOSEM 2026** (To appear)
2. **From Cool Demos to Production-Ready FMware: Core Challenges and a Technology Roadmap**
Gopi Krishnan Rajbahadur, Gustavo A. Oliva, Dayi Lin, **Jiho Shin**, Ahmed E. Hassan, **TOSEM 2026**
3. **Surveying the Benchmarking Landscape of Large Language Models in Code Intelligence**
M Abdollahi, R Zhang, NS Harzevili, **Jiho Shin**, S Wang, H Hemmati, **TOSEM 2026**
4. **Toward Automated Validation of Language Model Synthesized Test Cases using Semantic Entropy**
H Taherkhani, **Jiho Shin**, MA Tahir, MRH Misu, VS Gattani, H Hemmati, **TSE 2026**
5. **Assessing Evaluation Metrics for Neural Test Oracle Generation**
Jiho Shin, H Hemmati, MS Wei, S Wang, **TSE 2024 (ICSE-JF'25)**
6. **The Good, the Bad, and the Missing: Neural Code Generation for Machine Learning Tasks**
Jiho Shin, MS Wei, JJ Wang, L Shi, S Wang, **TOSEM 2023**
7. **A Survey of Automatic Code Generation from Natural Language**
Jiho Shin and JC Nam, **JIPS 2021**

Preprints

1. **Trajectory-Aware Benchmark Subset Selection for Cost-Efficient SWE-Agent Regression Testing**
Mahmoud Ayyad, Zehao Wang, **Jiho Shin**, Ying Zou, Bram Adams, arXiv 2026 (Under Review)
2. **NEUROTSTGEN: Neuro-Symbolic Guided Test Generation with Large Language Models**
Ruixin Zhang, **Jiho Shin**, Hung Viet Pham, Song Wang, arXiv 2026 (Under Review)
3. **A Large-Scale Empirical Study of Quality Assurance Practices and Gaps in AI Agents**
Wuyang Dai, Moses Openja, **Jiho Shin**, Hung Viet Pham, Song Wang, arXiv 2026 (Under Review)
4. **Testing Large Vision-Language Models: A Systematic Review**
M Openja, W Dai, **Jiho Shin**, N Akbarpour, C Davies, Belle, H Pham, G Uddin, S Wang, HAL 2026 (Under Review)
5. **BloomAPR: A Bloom's Taxonomy-based Framework for Assessing LLM-Powered APR Solutions**
Y Ma, **Jiho Shin**, LD Silva, ZM Jiang, S Wang, F Khomh, SH Tan, arXiv 2025 (Under Review, MajRev in TOSEM)

Invited Talks

Handong Global University - Delivered a seminar on rethinking AI for unit test generation	Mar 2026
Handong Global University - Delivered a talk on neuralware to promptware: unit test generation	Jan 2026
Korea University - Delivered a talk on neuralware to promptware: unit test generation	Jan 2026
Handong Global University - Invited as one of the alumni panelists at CEST (CSEE undergraduate MT)	Mar 2025
Handong Global University - Delivered a talk about different code generation methods in software engineering	Jan 2024

Awards/Grants/Funds

Distinguished Reviewer Award - Alware'26 distinguished reviewer award for Research Track .	Jul 2026
Best Benchmark & Dataset Award - Alware'26 Best Benchmark & Dataset Award.	Jul 2026
Outstanding Thesis Award - PhD dissertation awarded by the EECS Department at York University.	Jun 2026
Mitacs Business Strategy Internship - Funding for working with HGS (15K CAD).	Oct 2024
YorkU Academic Excellence Fund - Travel support for graduate research-related trips (2K CAD).	Jan 2024
APSEC - Distinguished Paper Award from APSEC'23 technical track.	Dec 2023
YorkU Graduate Fellowship - Paid research program (38K CAD per year for 4 years).	Sep 2021
KCSE - Distinguished Paper Award from KCSE'20 short paper track.	Feb 2020

Academic Services

Organizing Committee

- [ICSME'27] [Web Chair](#), 43rd IEEE International Conference on Software Maintenance and Evolution
[ICST'24] [Web Chair](#), 17th IEEE International Conference on Software Testing, Verification and Validation.

Student Volunteer

- [ICSE'25] [SV](#), 50th IEEE/ACM International Conference on Software Engineering.
[ICST'24] [SV](#), 17th IEEE International Conference on Software Testing, Verification and Validation.

PC

- [ASE'26] [PC](#), IEEE/ACM International Conference on Automated Software Engineering, **Research Paper Track**.
[ICST'26] [PC](#), IEEE International Conference on Software Testing, Verification and Validation, **Research Paper Track**.
[MSR'26] [PC](#), IEEE/ACM International Conference on Mining Software Repositories, **Mining Challenge Track**.
[FSE'26] [PC](#), ACM Foundations of Software Engineering, **Artifact Track**.
[AIGC'25] [PC](#), International Conference on AI-Generated Content, **Research Paper Track**.
[COMPSAC-SETA'25] [PC](#), IEEE Symposium on Software Engineering Technologies & Applications.
[MSR'25] [Junior PC](#), Junior PC for the IEEE/ACM MSR'25, **Research Paper Track** (accept%: 111/262=42.4%).
[ICSE'25] [Shadow PC](#), Shadow PC for the IEEE/ACM ICSE'25, **Research Paper Track** (accept%: 50/299=16.7%).

Reviewer

- [TSE] [Reviewer](#), IEEE Transactions on Software Engineering.
[TOSEM] [Reviewer](#), ACM Transactions on Software Engineering and Methodology.
[JSS] [Reviewer](#), ScienceDirect Journal of Systems and Software.
[EMSE] [Reviewer](#), Springer Nature Empirical Software Engineering.
[STVR] [Reviewer](#), Wiley Software Testing, Verification and Reliability.
[IEEE Software] [Reviewer](#), IEEE Software.
[ESWA] [Reviewer](#), ScienceDirect Expert Systems with Applications.
[PeerJ-CS] [Reviewer](#), PeerJ Computer Science.